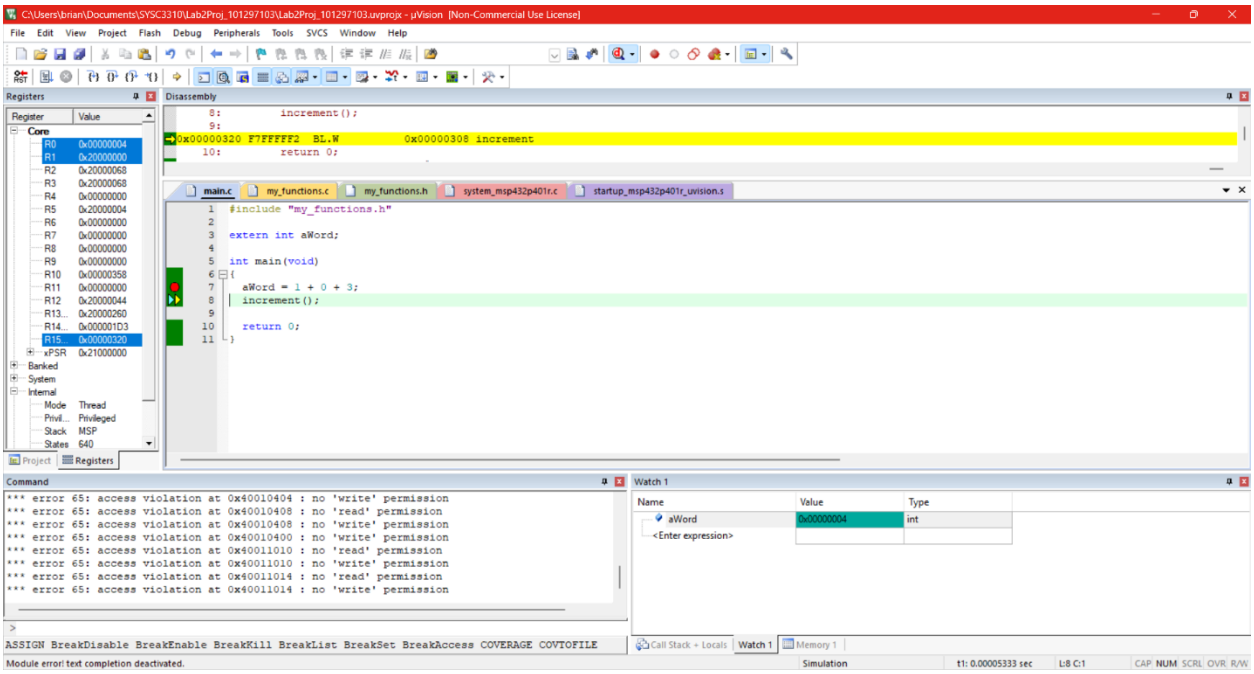
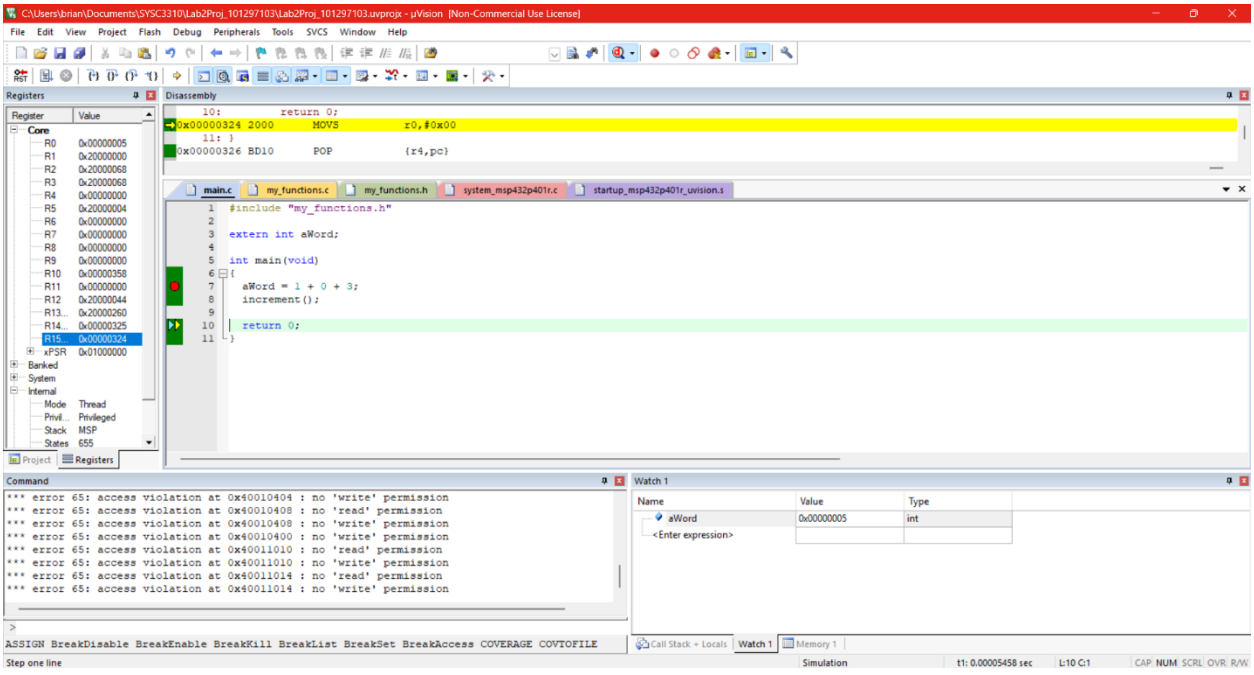
For Part 2:

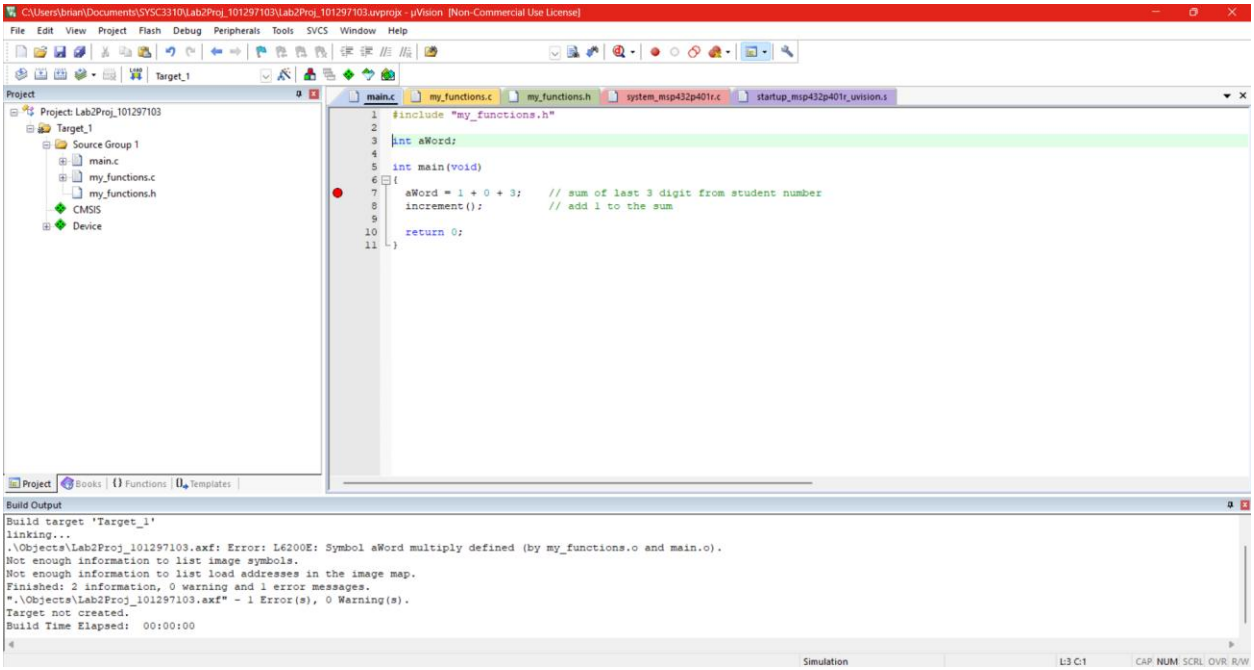
a. Screenshots of the Keil IDE showing the debugging view that shows the variable x before and after calling the function “increment”





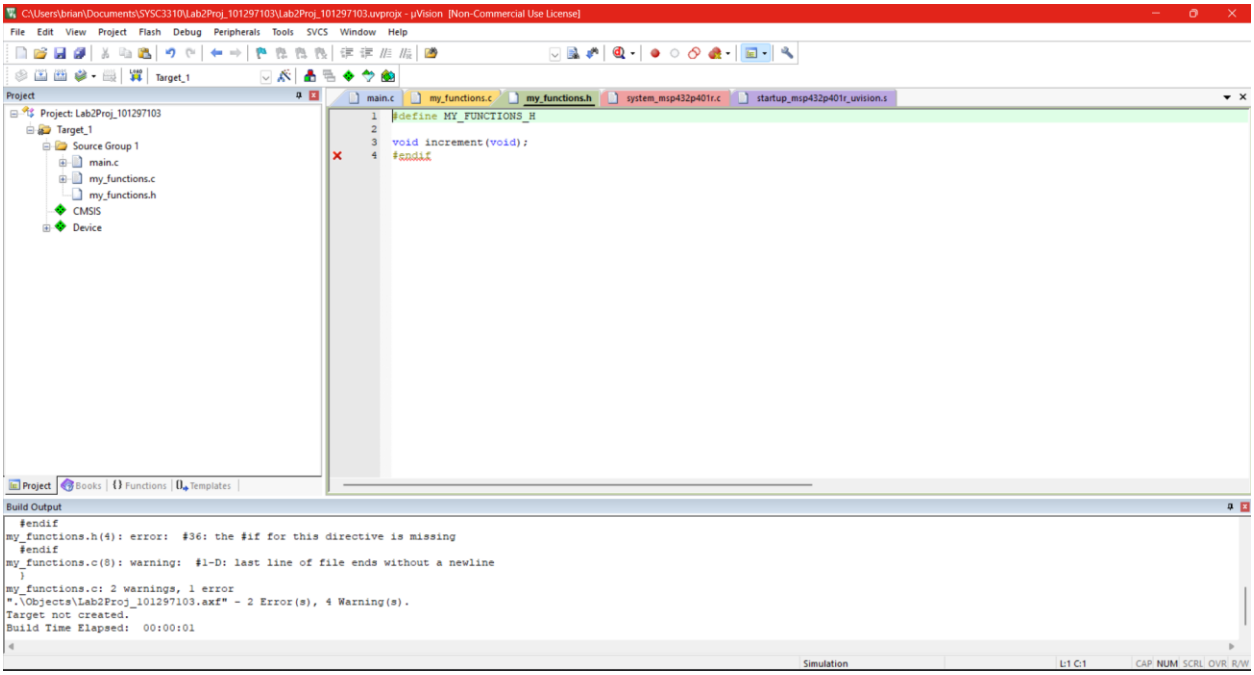
b. Brief description of the outcome of the following tests (show screenshots if necessary)

i. Remove the keyword “extern”



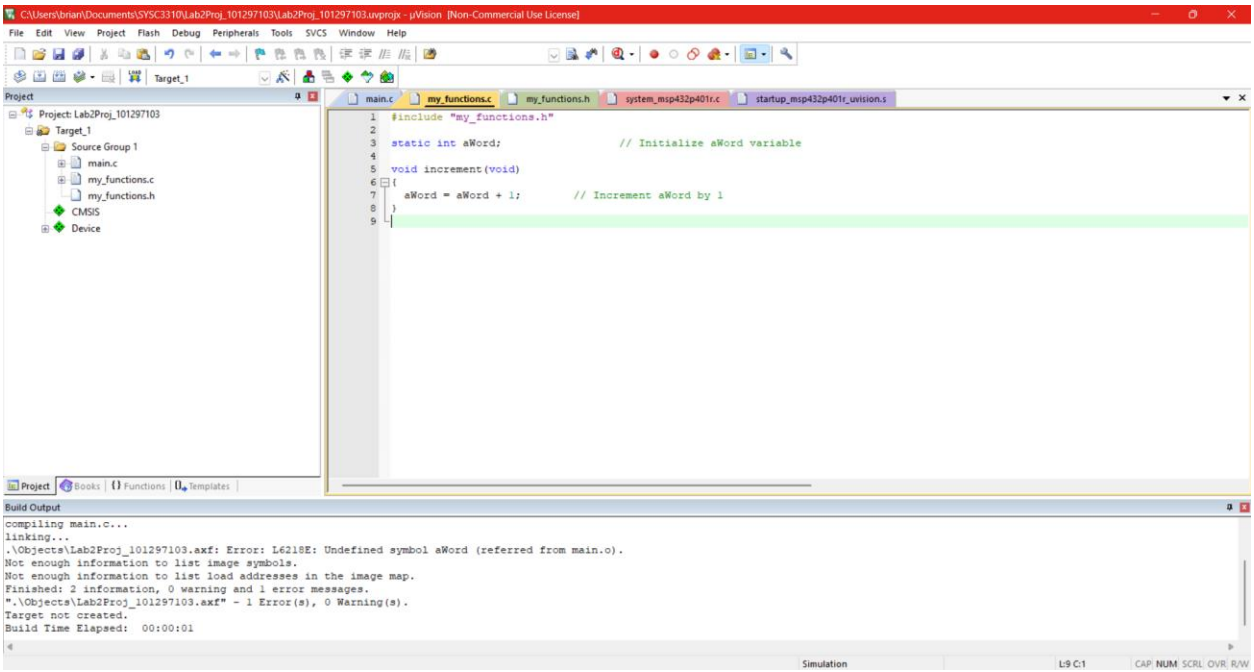
Removing the extern keyword caused a compilation error because aWord is defined in my_function.c, and main.c needs extern to indicate that the variable is defined in another source file.

ii. Removing #ifndef MY_FUNCTION_H



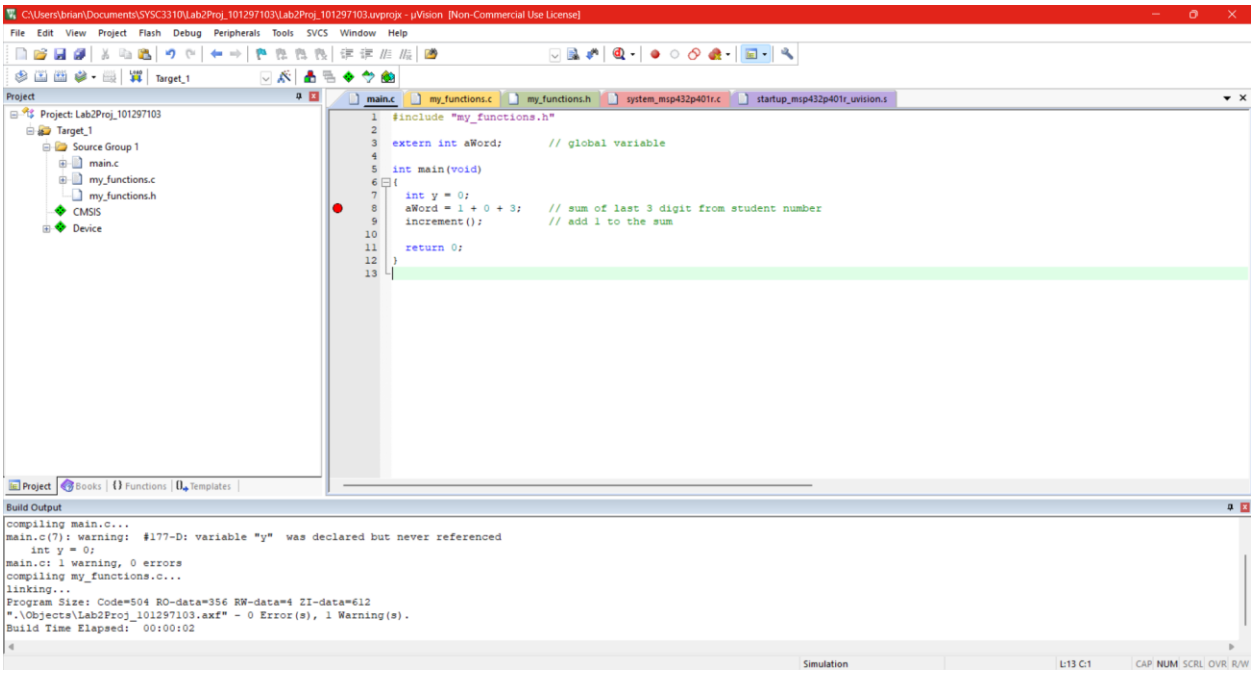
Removing #ifndef MY_FUNCTION_H caused a compilation error. Keil reported that the #endif directive was missing its corresponding #if directive. This happened because #endif is used to close the conditional directive started by #ifndef.

iii. Adding the keyword “static” before the definition of the variable x



Adding static before aWord caused a linking error. Keil said that aWord was undefined in main.c. This happened because static makes aWord available only in my_functions.c, so main.c cannot access it with extern.

iv. Define a local int variable “y” inside your main function and initialize it to zero



A local variable y was declared inside main() and initialized to zero. There are no compilation error, but keil produced a warning because y was declared but was never used.

For Part 3:

a. Screenshots of the Keil IDE showing the debugging view that shows the variables before and after running steps 5. and 10.

